

Research Paper

Linking uplift and mineralisation at the Mount Novit Zn-Pb-Ag Deposit, Northern Australia: Evidence from geology, U-Pb geochronology and sphalerite geochemistry

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ARTICLE INFO

Article history:

Received 13 September 2021

Revised 9 November 2021

Accepted 30 December 2021

Available online 01 January 2022

Handling Editor: K. Szilas

Keywords:

Mount Isa

Mount Novit

Zn-Pb-Ag mineralisation

Monazite

Apatite

Geochronology

ABSTRACT

The subeconomic Mount Novit Zn-Pb-Ag deposit is located approximately 20 km south of Mount Isa, Queensland. In contrast to the nearby Mount Isa, Hilton and George Fisher Zn-Pb-Ag deposits, mineralisation at Mount Novit is situated to the west of the regional-scale Mount Isa Fault and is hosted in the Moondarra Siltstone as opposed to the Urquhart Shale. Lower-grade (<4 wt.% Zn + Pb) Zn-Pb-Ag mineralisation primarily replaces pre-existing carbonate alteration and veining and consists of pyrrhotite, pyrite and sphalerite with lesser galena. Higher-grade (>10 wt.% Zn + Pb) mineralisation occurs as a matrix supported breccia dominated by sphalerite and pyrrhotite with galena, pyrite, and magnetite. In-situ U-Pb geochronology was completed on apatite and two textural varieties of monazite. Fine-grained (<50 µm) subhedral to anhedral monazite is located within highly foliated biotite alteration directly adjacent Zn-Pb-Ag mineralisation and yields a mean weighted $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1527 ± 18 Ma (MSWD = 1.06). This age is consistent with the formation of highly foliated biotite alteration during D_3 deformation of the Isan Orogeny. Apatite from the same fabric yields a lower intercept age of 1443 ± 29 Ma (MSWD = 1.30). Consistent with previous studies, this age is interpreted to represent the age of a major thrusting event along the Mount Isa Fault that resulted in the cooling of the Mount Novit area below ~ 375 °C. Coarse-grained monazite is coeval with Zn-Pb-Ag mineralisation and yields a mean weighted $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1457 ± 11 Ma (MSWD = 0.28). Sphalerite from Mount Novit has low concentrations (<1 ppm) of Ge and Ga and a relatively high concentration of In (5 to >10 ppm), possibly reflecting the leaching of the metals from an underlying basement unit. The GGIMFis geothermometer (Frenzel et al., 2016) produced a mean formation temperature of 345 ± 52 °C. The timing and temperature of Zn-Pb-Ag mineralisation is consistent with the age and cooling temperature of apatite presented in this study. Based on these correlations, we suggest that Zn-Pb-Ag mineralisation at Mount Novit was emplaced during an episode of major thrusting along the Mount Isa Fault, with the precipitation of Zn-Pb-Ag mineralisation driven by the cooling of the Mount Novit area below ~ 375 °C. A key implication of this study is a new model for synorogenic Zn-Pb-Ag mineralisation to the south of Mount Isa, which contrasts with the widely accepted regional-scale syngenetic metallogenic model.

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1. Introduction

The Mount Isa region, northwest Queensland, is a globally significant Zn-Pb-Ag metallogenic province containing numerous world-class sediment-hosted Zn-Pb-Ag deposits. The Mount Isa (Cu-Zn-Pb-Ag), George Fisher (Zn-Pb-Ag) and Hilton (Zn-Pb-Ag) deposits are all situated within 15 km of each other and share mul-

tiply distinct characteristics. These deposits are all hosted by the Urquhart Shale, and are all situated to the east of the regional-scale Mount Isa Fault, with Zn-Pb-Ag mineralisation occurring as multiple conformable and stacked stratabound lenses (Mathias and Clark, 1975; Perkins and Bell, 1998; Chapman, 2004). Due to the size and economic importance of these deposits, their metallogenic models are frequently used to guide regional exploration. Syngenetic or syn-inversion metallogenic models have been widely accepted for these deposits and involve the formation of the Zn-Pb-Ag mineralisation during, or slightly after diagenesis of

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the host lithology at ca. 1650 Ma (Valenta, 1994a; Carr et al., 2004; Chapman, 2004; Large et al., 2005; Rieger et al., 2020). However, epigenetic and syn-deformation metallogenic models have also been proposed for these deposits and involves the introduction of Zn-Pb-Ag mineralisation during the late stages of the 1610–1500 Ma Isan Orogeny (Perkins, 1997; Perkins and Bell, 1998; Davis, 2004; Cave et al., 2020a).

The Mount Novit Zn-Pb-Ag deposit is a small accumulation of subeconomic mineralisation (estimated metal content: 1.17 Mt Zn, 0.82 Mt Pb and 0.001 Mt Ag; Stoker, 1981) located approximately 20 km south of Mount Isa (Fig. 1). Zn-Pb-Ag mineralisation at Mount Novit strikes N–S and crops out as a Fe-rich gossan over a distance of approximately 6 km (Fig. 1; Perkins, 1996). Due to its subeconomic grade and rather limited exposure, Zn-Pb-Ag mineralisation at Mount Novit has been the subject of little geological study (McGoldrick, 1996; Perkins, 1996; Carr et al., 2001). Zn-Pb-Ag mineralisation at Mount Novit differs from the nearby Mount Isa, George Fisher and Hilton deposits as it is hosted by an overturned sequence of the stratigraphically older Moondarra Siltstone and occurs to the west rather than the east of the Mount Isa Fault (Fig. 2; Perkins, 1996). These key differences have potential implications for the local- and regional-scale Zn-Pb-Ag metallogenic model, warranting further investigation.

In this study, two sections of drill core that intersect separate mineralised horizons at the Mount Novit Zn-Pb-Ag deposit are sampled and logged in detail. Petrography is combined with in-situ monazite and apatite U–Pb geochronology in order to establish a relationship between metamorphism, cooling, and Zn-Pb-Ag

mineralisation. The geochemistry of sphalerite is used to assess the temperature of Zn-Pb-Ag formation and to characterise potential metal sources. A metallogenic model for the Mount Novit Zn-Pb-Ag deposit is proposed that incorporates the textural, geochronological, and physiochemical constraints presented in this study.

2. Geological setting

2.1. Regional geology

The Paleoproterozoic Mount Isa Inlier consists of three major structural belts: (1) Western Fold Belt; (2) Kalkadoon–Leichardt Belt; (3) Eastern Fold Belt, which are interpreted to be separated by major N–S striking fault zones (Blake, 1987). The Western Fold Belt is host to three major superbasins: (1) Leichhardt Superbasin (1800–1750 Ma); (2) Calvert Superbasin (1740–1690 Ma); and (3) Isa Superbasin (1670–1580 Ma), which were terminated by the onset of the 1610–1500 Ma Isan Orogeny (Bell, 1983; Page and Bell, 1986; Page et al., 2000; Scott et al., 2000). In the Mount Novit region, the Calvert and Isa Superbasins crop out extensively and are composed of the various volcanic and sedimentary packages that make up the Gun, Loretta, River, Term, Lawn, Wide and Doom Supersequences (Page et al., 2000; Southgate et al., 2013). The Prize and Gun Supersequences are dominant in the Mount Novit area and consist of (in stratigraphic order) the Surprise Creek Formation, Moondarra Siltstone, Breakaway Shale and the Native Bee Siltstone (Domagala et al., 2000; Southgate et al., 2013).

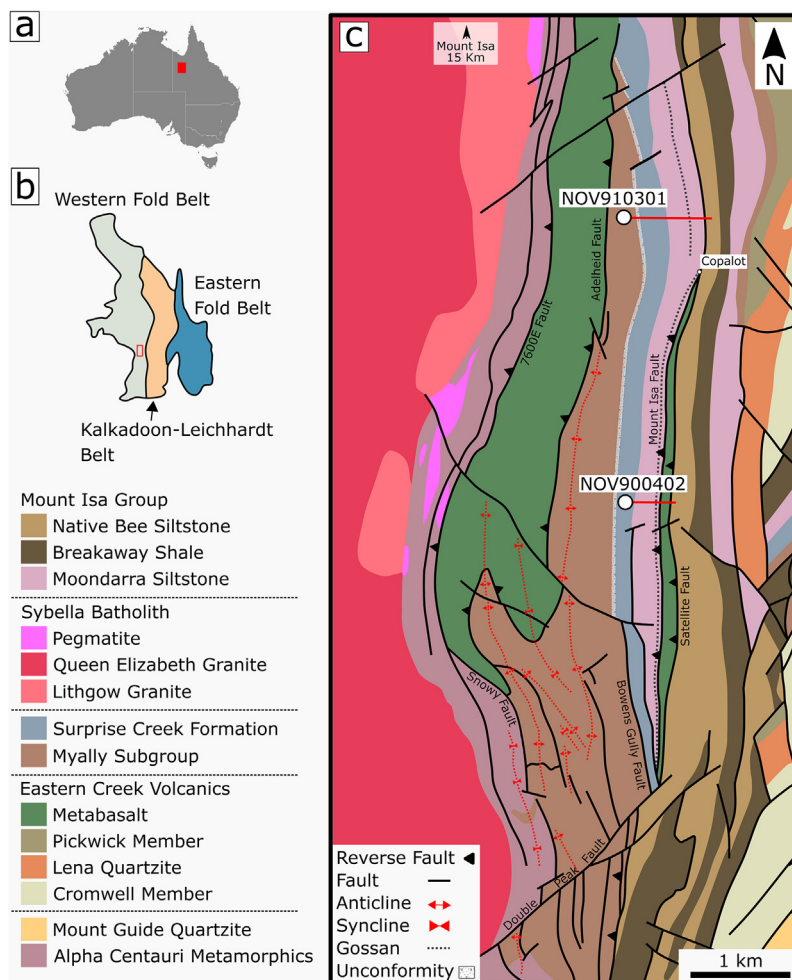


Fig. 1. Geological map of the Mount Novit area modified from the Geological Survey of Queensland (2020). (a) Location of the Mount Isa Inlier with reference to Australia. (b) Location of the Mount Novit area with reference to the Mount Isa Inlier. (c) Simplified geological map of the Mount Novit area.

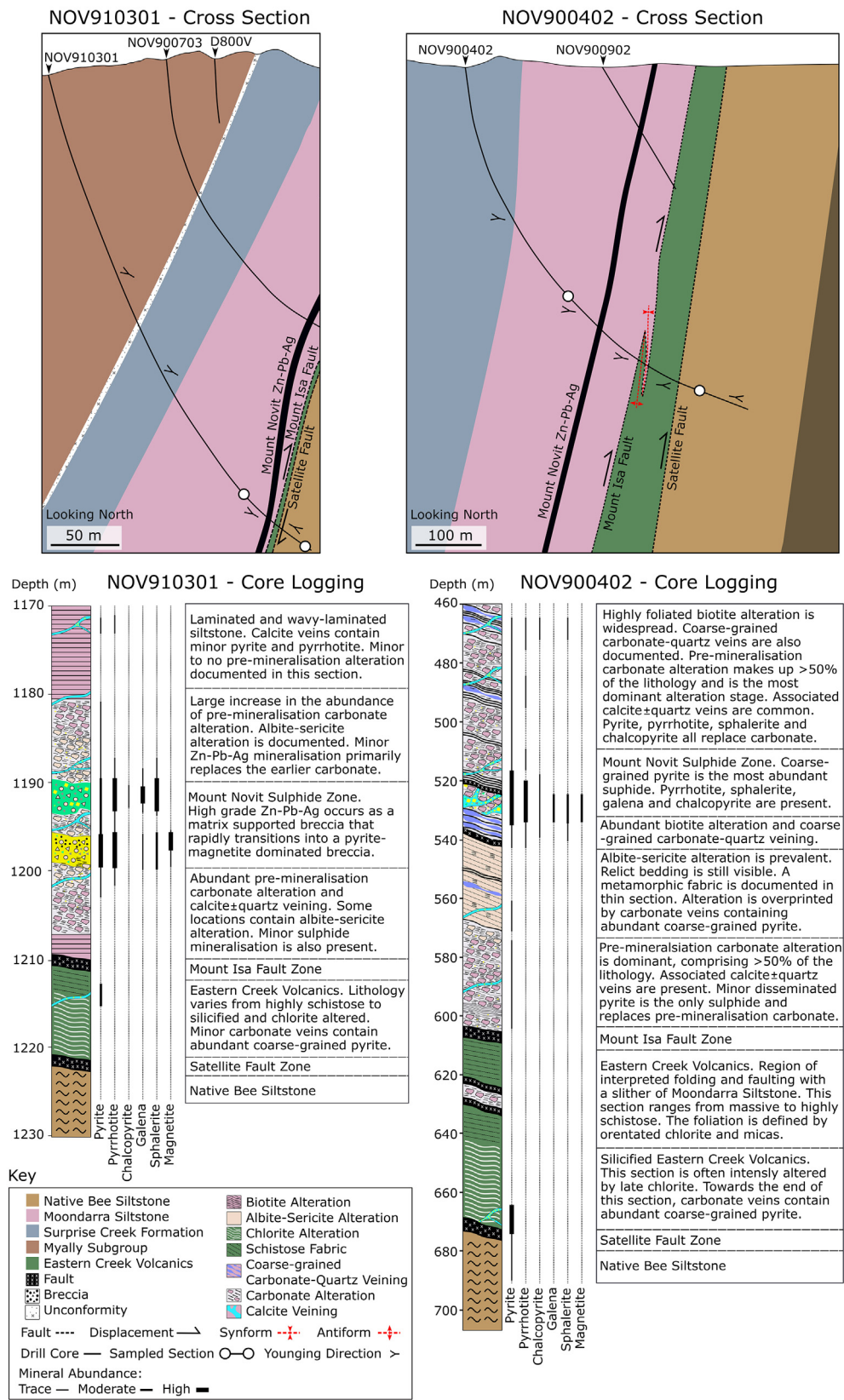


Fig. 2. Cross sections of the Mount Novit Zn-Pb-Ag prospect. The logged lithologies, corresponding mineral abundances and a brief summary of the logged sections are shown. The location of the core logged and sampled in this study is shown in Fig. 1.

Deformation pre-dating the Isan Orogeny has previously been recorded in the Mount Isa region (Derrick, 1982). Smith (1969) was the first to recognise syn-Mount Isa Group penecontemporaneous faulting in the region, but its significance has been widely debated. The Mount Isa Inlier experienced multiple stages of shortening during the polyphase (1610–1500 Ma) Isan Orogeny (Bell, 1983; Betts et al., 2006). This study utilizes the D_1 – D_2 – D_3 deformation scheme of the Isan Orogeny most widely used in the literature (Bell, 1983; Winsor, 1986; MacCready, 2006a). Early N–S directed shortening during D_1 deformation resulted in the inversion of pre-existing E–W rift structures (Bell, 1983; Winsor, 1986). The configuration of these pre-existing structures was likely controlled by an earlier basin and dome folding event that locally developed overturning prior to the deposition of the Surprise Creek Formation (Proffett, 1990). D_1 deformation also facilitated the development of E–W trending upright folds and the formation of a lineation sub-parallel to bedding (Bell, 1983; Winsor, 1986; Betts et al., 2006; MacCready, 2006b). Although the timing of this early event is poorly constrained, regions of the Sybella Granite interpreted to have been affected by D_1 deformation produce a whole rock Rb–Sr age of 1610 ± 10 Ma (Page and Bell, 1986). D_1 -related fabrics are largely absent in the Mount Novit region, with pre- S_2 fabrics only preserved in the strain shadow of cordierite porphyroblasts (Connors and Lister, 1995). In the Mount Novit area, D_2 is the most pervasive stage of deformation and is synchronous with upper amphibolite facies metamorphism (Connors et al., 1992). W–E directed shortening during D_2 deformation produced regional-scale N–S trending folds and the dominant S_2 fabric defined by the alignment of micas in pelitic units and the alignment of chlorite in mafic units (Connors and Lister, 1995). D_2 is followed by a subsequent event ($D_{2.5}$) that is restricted to localised areas affected by D_2 deformation (Bell and Hickey, 1998). $D_{2.5}$ (termed D_3 by Davis (2004) and Cave et al. (2020a)) produced highly asymmetric deflections in S_2 fabric and bedding with top to the east movement (Bell and Hickey, 1998). In the Mount Novit area, brittle-ductile D_3 deformation occurred during retrograde metamorphic conditions producing major faulting and a spaced cleavage associated with upright, predominantly non-cylindrical folds that trend NW–SE (Connors and Lister, 1995; MacCready, 2006a). Using whole-rock Rb–Sr, early studies dated D_3 deformation at 1510 ± 13 Ma (Page and Bell, 1986). In the Mount Novit area, a post D_3 pegmatite has been dated at 1480 ± 14 Ma (U–Pb zircon), providing a maximum age of regional D_3 deformation (Connors and Page, 1995). In isolated areas, late folds and associated cleavages (D_{4-5} corresponding to F_{4-5} and S_{4-5}) have been documented in the Mount Novit region and are interpreted to have formed in the same kinematic framework as D_3 during an ongoing period of prolonged deformation (Connors et al., 1992; Connors and Lister, 1995).

2.2. Deposit geology

Zn–Pb–Ag mineralisation at Mount Novit is hosted by an overturned sequence of the Moondarra Siltstone (minimum age 1663 ± 3 Ma; Page et al., 2000) that sits three lithostratigraphic units below the Urquhart Shale (Domagala et al., 2000). Sulphide mineralisation at Mount Novit crops out as a gossanous ridge and occurs as two discontinuous N–S orientated sulphide horizons that broadly parallel the Mount Isa Fault (Fig. 1; Perkins, 1996). The northernmost Zn–Pb–Ag horizon is bound by the intersection of the Mount Isa Fault and an unnamed E–W trending fault to the north, and continuous to the south until it terminates ~150 m north of the Copalot shaft (Fig. 1). The southern Zn–Pb–Ag horizon is positioned ~130 m east relative to the northern mineralised horizon. The southernmost Zn–Pb–Ag horizon trends south from the Copalot shaft where it terminates at the intersection of the Mount Isa Fault and the Double Peak Fault (Fig. 1). While drill core in the area

intersects sulphide mineralisation over sections >30 m, intersections containing Zn–Pb–Ag mineralisation have a maximum true width of 4.1 m (Perkins, 1996). At depth, the Mount Novit Zn–Pb–Ag horizon sits parallel to the adjacent Mount Isa and Satellite Faults (Fig. 2). The Mount Isa Fault dips 60–70° to the west and places greenschist- to amphibolite-facies, eastward younging rocks of the Mount Isa Group against westward younging, lower greenschist-facies rocks of the same unit (Fig. 2; Connors et al., 1992; MacCready, 2006b). In the immediate vicinity of this study, the Mount Isa Fault positions the Moondarra Siltstone against the Eastern Creek Volcanics (Fig. 2). Similarly, the Satellite Fault dips steeply westward, positioning the Eastern Creek Volcanics against the westward younging Native Bee Siltstone (Fig. 2). A S_2 fabric has been identified along silicified portions of the Satellite Fault, leading to its interpretation as a pre- S_2 low-angle thrust fault (MacCready, 2006a). In drill core NOV900402, a small synform-antiform pair has been interpreted within a zone of intense deformation associated with the Mount Isa Fault (Fig. 2). These folds are interpreted to have formed during the reverse reactivation of the Mount Isa Fault during ongoing D_3 deformation of the Isan Orogeny.

3. Methods

3.1. Monazite U–Pb geochronology

Monazite U–Pb geochronology and trace element concentrations were acquired using a RESOLUTION 193 nm excimer laser ablation system coupled with an Agilent 7900 quadrupole ICP–MS at Adelaide Microscopy. Ablations were conducted in an atmosphere of He (0.35 L/min) that was mixed with Ar (1.06 L/min) upon leaving the ablation cell. The material then passes through a pulse-homogenization device (Squid, Laurin Technic Inc.) before being introduced to the ICP torch. The repetition rate of the laser was set at 5 Hz with a consistent fluence of 2.0 J/cm². Pre-ablation consisted of 5 pulses followed by a 20 s delay to allow adequate cell wash out for the subsequent analyses. Data acquisition consisted of 30 s of background collection, followed by 30 s of ablation. The isotopes analysed include: ²⁷Al, ³¹P, ⁴³Ca, ⁸⁹Y, ⁹⁰Zr, ¹³⁹La, ¹⁴⁰Ce, ¹⁴¹Pr, ¹⁴⁶Nd, ¹⁴⁷Sm, ¹⁵³Eu, ¹⁵⁷Gd, ¹⁵⁹Tb, ¹⁶³Dy, ¹⁶⁵Ho, ¹⁶⁶Er, ¹⁶⁹Tm, ¹⁷²Yb, ¹⁷⁵Lu, ²⁰²Hg, ²⁰⁴Pb, ²⁰⁶Pb, ²⁰⁷Pb, ²⁰⁸Pb, ²³²Th and ²³⁸U. A 13 µm spot was used for monazite situated in highly foliated biotite-rich alteration zones and corresponding standard analyses. A 19 µm spot was used on monazite associated with Zn–Pb–Ag mineralisation and corresponding standard analyses. LA–ICP–MS data was processed using the VisualAge Data Reduction Scheme (DRS) (Petrus and Kamber, 2012) in Iolite (Paton et al., 2011). The MADEL monazite standard (TIMS ²⁰⁶Pb/²³⁸U age = 518.37 ± 0.99 Ma; ²⁰⁷Pb/²⁰⁶Pb = 490.7 ± 1.1 Ma; Payne et al., 2008) was used as a primary calibration standard using a 20:2 standard sample bracketing approach to correct for mass bias and elemental fractionation of the U–Pb ratios. Trace element concentrations were calculated using the Trace Element_IS DRS in Iolite 3.32 (Paton et al., 2011). NIST610 glass was used as the primary standard for the quantification of trace elements using an internal standard of 20.00 wt.% Ce. The results were then normalised to 100 wt.% assuming a stoichiometric monazite composition. Data accuracy was monitored by the analysis of the in-house secondary monazite standards 94–222 (²⁰⁶Pb/²³⁸U age = 450.2 ± 3.4 Ma) (Maidment, 2005) and Ambat (²⁰⁶Pb/²³⁸U age = ca. 520 Ma). The resulting ²⁰⁶Pb/²³⁸U mean weighted ages of the analysed standards 94–222 are 450 ± 2 Ma ($n = 24/24$); and Ambat = 520 ± 2 Ma ($n = 24/24$). U–Pb ages were plotted and calculated using IsoplotR (Vermeesch, 2018). Errors of all mean weighted ages are quoted at the 95% confidence interval.

3.2. Apatite U–Pb geochronology

Apatite U–Pb geochronology was performed using a RESOLUTION-LR 193 nm excimer laser coupled to an Agilent 7900 ICP-MS system at Adelaide Microscopy. The analytical procedures and instrumental settings used in this study are outlined in Cave et al. (2018). Data reduction was performed using the VisualAge_Ucompbine data reduction scheme (Chew et al., 2014) in Iolite 3.32 (Paton et al., 2011). Madagascar apatite (U–Pb TIMS = 473.5 ± 0.7 Ma; Chew et al., 2014; Thomson et al., 2012) was used as the primary calibration standard to correct for mass bias and elemental fractionation of the U–Pb ratios. McClure Mountain (U–Pb TIMS age of 523.51 ± 1.47 Ma) (Schoene and Bowring, 2006) and 401 (530.3 ± 1.5 Ma; Thompson et al., 2016) apatite were used as secondary standards to perform accuracy checks. Trace element concentrations were calculated using the Trace Element_IS DRS in Iolite (Paton et al., 2011). NIST610 glass was used as the primary standard for the quantification trace elements with an internal standard value of 39.36 wt.% Ca. An unanchored linear regression was used to calculate the age of the analysed apatite in this study as it does not make assumptions regarding the initial $^{207}\text{Pb}/^{206}\text{Pb}$ composition. The data collected over the analytical session yields a ^{207}Pb corrected age of 531 ± 6 Ma for the McClure Mountain apatite and a ^{207}Pb corrected age of 528 ± 5 Ma for the 401 apatite. Data was plotted and U–Pb ages were calculated using IsoplotR (Vermeesch, 2018). Errors associated with the U–Pb intercept age are quoted at the 95% confidence interval.

3.3. Sphalerite trace elements

Sphalerite LA-ICP-MS spot analyses was acquired using a RESOLUTION 193 nm excimer laser ablation system coupled with an Agilent 7900 quadrupole ICP-MS at Adelaide Microscopy. Ablation spot size ranged from 10–70 μm . The repetition rate of the laser was 5 Hz with a consistent fluence of 3.0 J/cm². Pre-ablation consisted of 5 pulses, followed by a 20 s delay to allow adequate cell wash out before the subsequent analyses. Data collection consisted of 30 s background collection, followed by 40 s of ablation. The isotopes analysed include: ^{29}Si , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{65}Cu , ^{66}Zn , ^{69}Ga , ^{71}Ga , ^{72}Ge , ^{73}Ge , ^{75}As , ^{77}Se , ^{95}Mo , ^{109}Ag , ^{111}Cd , ^{113}In , ^{115}In , ^{118}Sn , ^{121}Sb , ^{125}Te , ^{182}W , ^{197}Au , ^{201}Hg , ^{205}Tl , ^{206}Pb , ^{207}Pb , and ^{209}Bi . The GSD-1G (USGS) and STDGL3 (CODES, University of Tasmania; Danyushevsky et al., 2011) standards were analysed using a 70 μm diameter spot size with a 25:2 standard sample bracketing approach. Analyses were processed using LADR data reduction software (Norris and Danyushevsky, 2018) following the methodology of Cave et al. (2020b). Each analysis was checked for mineral inclusions by inspecting the time-resolved profile of each individual isotope during data processing. Data with obvious inclusions were rejected, and inclusion-free portions of the time-resolved profile were selected. The STDGL3 standard was used as the primary standard to correct for instrument drift and mass bias, with GSD-1G used as a secondary standard. For the quantification of sphalerite, the measured isotopes were normalised to a total of 100 wt.% assuming mono-sulphide stoichiometry. Where multiple isotopes were measured, the concentration of ^{115}In , ^{208}Pb , ^{69}Ga and ^{72}Ge are preferentially quoted.

4. Results

4.1. Petrology and core logging

This study examines the geological characteristics of two sections of drill core located approximately 2 km apart that intersect

the Mount Novit Zn–Pb–Ag horizons (Figs. 1 and 2). Drill core NOV910301 intersects the northern Zn–Pb–Ag horizon while drill core NOV900402 intersects the southern Zn–Pb–Ag horizon (Fig. 1). In addition to the eight thin sections produced from collected samples, seven additional thin sections were sourced from The Queensland Museum and represent samples from Mount Novit used in Perkins (1996). In order of occurrence, the various lithologies and mineral assemblages documented in the studied core are described below:

4.1.1. Eastern Creek Volcanics

In both sections of the drill core logged in this study the Eastern Creek Volcanics are situated on the eastern side of the Mount Isa Fault (Fig. 2). Two distinct lithologies are documented in the Eastern Creek Volcanics. The first lithology is a highly foliated chlorite-rich schist that appears from 1210–1212 m in drill core NOV910301 and 608–620 m and 630–640 m in drill core NOV900402 (Fig. 2). In hand specimen, this lithology appears as a dark green highly foliated rock often containing calcite veins (Fig. 3a). Titanomagnetite, quartz, calcite, feldspar with minor apatite and pyrite make up the dominant mineral assemblage (Fig. 4a–b). These minerals are often rotated and enveloped by fabric-defining chlorite (Fig. 4a–b). Small (mm scale) calcite veins are often deformed and overprinted by the foliated chlorite. These textural relationships constrain the timing of calcite veining to either before or during early schist development. The second lithology is a highly silicified and chlorite altered portion of the Eastern Creek Volcanics (Fig. 3b). These rocks possess minor to no remnant fabrics and are composed predominantly of quartz (>95%). This lithology occurs from 1212–1220 m in drill core NOV910301 and 642–668 m in drill core NOV900402 (Fig. 2). The silicified portions of the Eastern Creek Volcanics occasionally contain coarse-grained quartz veins that are subsequently overprinted by late chlorite alteration, which can consume up to 20% of the remaining rock (Fig. 3b).

4.1.2. Moondarra Siltstone

The Moondarra Siltstone is the host of Zn–Pb–Ag mineralisation at the Mount Novit deposit. In the sections of core logged in this study the Moondarra Siltstone is composed of laminated, wavy-laminated and semi-massive metasilstones (Fig. 3c–e). Fine-grained muscovite and quartz dominate the siltstones mineralogy (Fig. 4c–d) with trace amounts of apatite, rutile, zircon, and monazite. A wavy heavy-mineral band consisting of medium-grained quartz, rutile and pyrite is visible in a sample collected at 1179 m from drill core NOV910301. Graphitic material occurs sporadically throughout the siltstone, and is often deformed, folded, and rotated, consistent with its formation prior to deformation. All samples analysed in this study have been variably metamorphosed, and commonly possess a prominent S_2 foliation (Fig. 4c–d). The S_2 fabric is predominantly defined by the orientation of fine-grained muscovite (Fig. 4c–d) and the trace minerals apatite, zircon and rutile.

4.1.3. Albite-sericite alteration

Albite-sericite alteration is documented from 1185–1208 m in drill core NOV910301 and from 538–570 m in drill core NOV900402 (Fig. 2). This lithology is consistent with the replacement of the original siltstone components by fine-grained albite and sericite with minor quartz, producing an often-homogeneous buff-coloured rock (Fig. 3f–g). Relict bedding is still observed in some specimens (Fig. 3g). In thin section, albite and sericite are often strongly orientated with S_2 , consistent with a metamorphic overprint. Albite-sericite alteration is overprinted by coarse-grained carbonate-quartz veining, carbonate alteration, and sulphide mineralisation (Fig. 3f–g). Carbonate alteration and veining

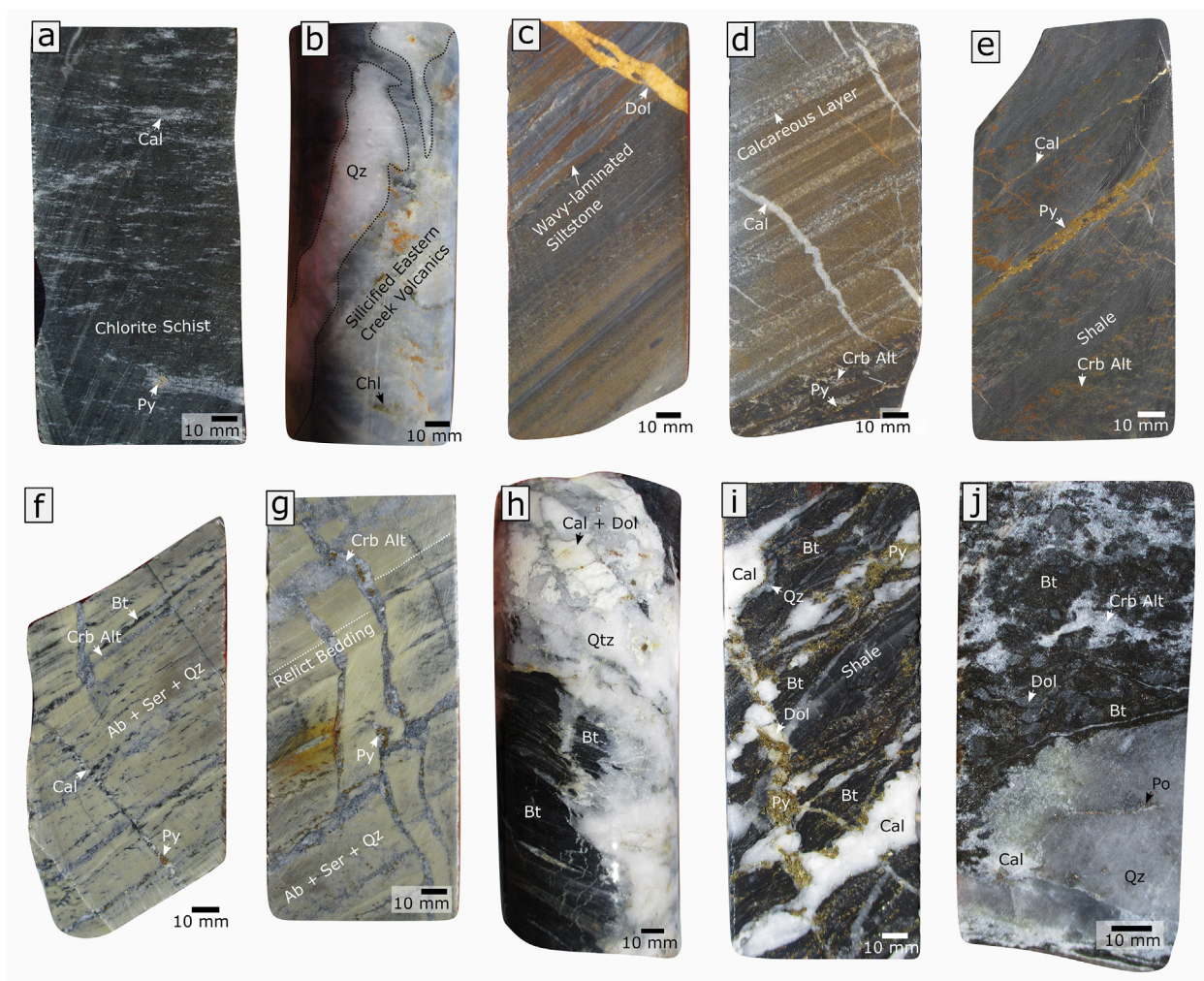


Fig. 3. Drill core samples of the host rocks and various stages of alteration and veining at the Mount Novit Zn-Pb-Ag deposit. Two distinct lithologies are present in the Eastern Creek Volcanics: (a) highly foliated chlorite schist and; (b) highly silicified portions. A late quartz vein is documented in (b). The Moondarra Siltstone host rock that includes: (c) wavy-laminated siltstone; (d) calcareous-layered siltstone and; (e) massive siltstone. (f, g) Siltstone that has undergone advanced albite-sericite alteration. Relict bedding is visible in (g). (h–j) Highly foliated biotite alteration overprinted by coarse-grained carbonate-quartz veining. Note the presence of minor boudins in (i) resulting from post-emplacement deformation. Pyrite is late and occurs along the grain boundaries and recrystallized portions of the coarse-grained quartz and calcite. Abbreviations: Ab = Albite, Bt = Biotite, Cal = Calcite, Chl = Chlorite, Crb Alt = Carbonate Alteration, Dol = Dolomite, Po = Pyrrhotite, Py = Pyrite, Qz = Quartz, Ser = Sericite.

occurs as both discordant and stratabound, often producing the appearance of a highly layered rock (Fig. 3f–g).

4.1.4. Highly foliated biotite alteration

A highly foliated biotite-rich assemblage occurs intermittently from 460–540 m in drill core NOV900402 (Figs. 2 and 3h–j). In quartz-rich psammitic units, biotite-rich alteration consumes up to 70% of the host siltstone. Individual biotite grains range from 100–250 μm in length and are often strongly foliated (Fig. 4e–g). Fabric-defining biotite wraps around and often penetrates dolomite porphyroclasts (Fig. 4e–g). In lower strain zones such as areas adjacent to the dolomite porphyroclasts, biotite displays decussate textures. Veins containing coeval calcite range from 0.1–1.0 mm in width and are developed parallel to the fabric-defining biotite (Fig. 4e). Medium-grained poikiloblastic pyrite is relatively uncommon and intergrown with biotite. Considering the orientation of the drill core, the biotite-dominated foliation dips $\sim 80^\circ$ west. In the vicinity of the Mount Isa Fault, S_2 and S_3 are often parallel and difficult to differentiate (Connors et al., 1992; Connors and Lister, 1995; MacCready, 2006a). Based on the U–Pb age of monazite situated in the highly foliated biotite alteration assemblage

(See Section 4.2), its formation during late D_3 deformation, and therefore representation of S_3 is favoured.

4.1.5. Coarse-grained carbonate-quartz veining

Coarse-grained carbonate-quartz veining is heterogeneously distributed throughout the sections of drill core analysed in this study (Figs. 2 and 3h–j). The region of the most intense carbonate-quartz veining occurs intermittently throughout a highly-fractured zone from 532–540 m in drill core NOV900402 (Fig. 2). Early coarse-grained calcite (80–90%) with lesser dolomite (10–20%) occur in veins ranging from 1 to >5 cm in width. Frequently, coarse-grained carbonate veins are overprinted by coarse-grained (1–10 cm width) quartz (Fig. 3h). These minerals are commonly constrained to a singular vein, consistent with a similar timing. Coarse-grained carbonate-quartz veins display evidence of significant post-emplacement deformation. In hand specimen, deformed veins appear as boudins often enveloped by highly foliated biotite alteration (Fig. 3i). In thin section, coarse-grained quartz displays mortar textures that have resulted from dynamic recrystallisation during deformation (Fig. 4h–i). The zones occupied by the fine-grained recrystallized quartz are often penetrated

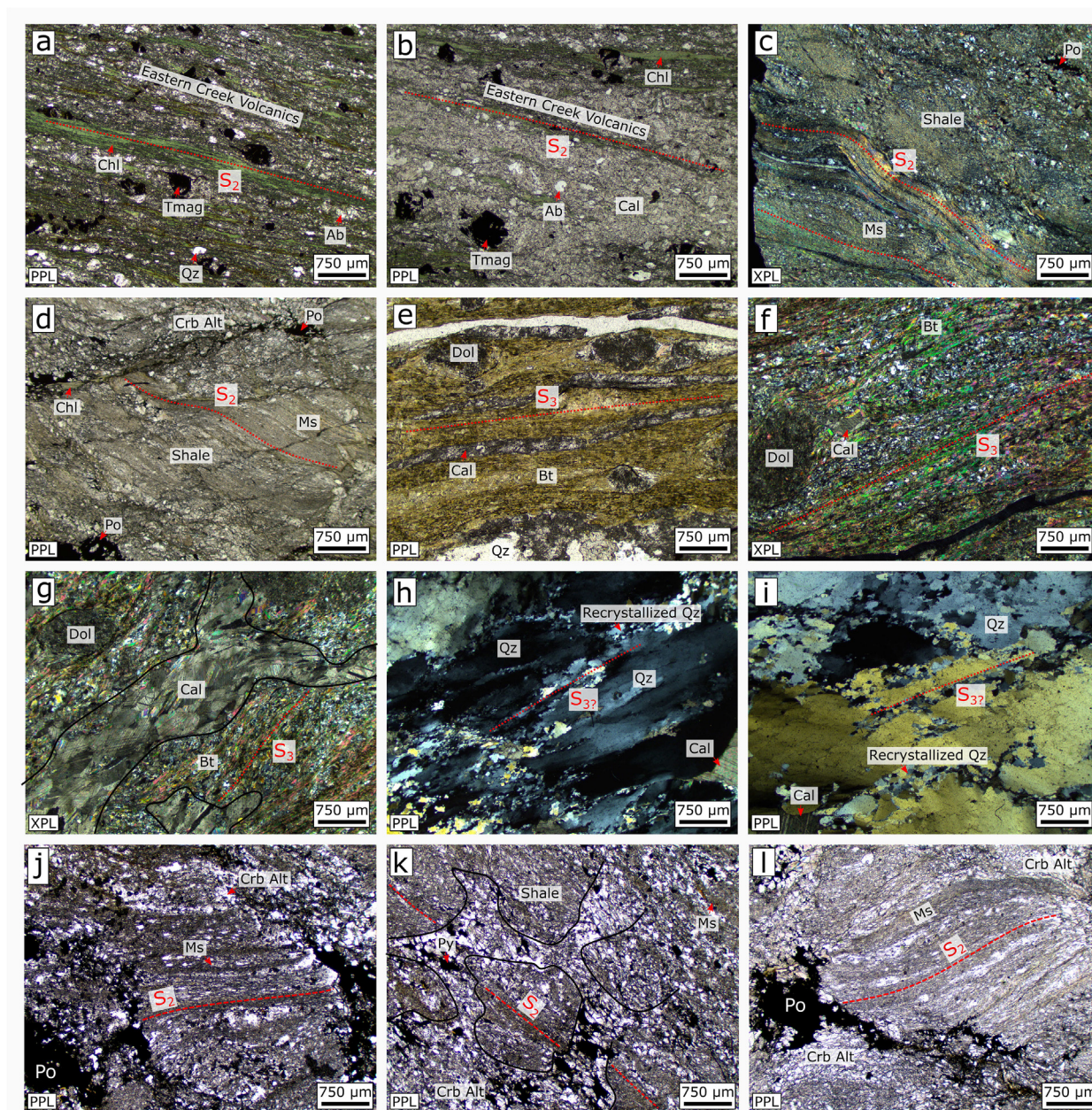


Fig. 4. Photomicrographs of the host rocks and multiple stages of alteration and veining at the Mount Novit Zn-Pb-Ag deposit. (a, b) Chloritic schist portion of the Eastern Creek Volcanics. Note calcite veining in (b) overprinted by fabric defining chlorite. (c, d) Metamorphosed siltstone host rocks with S_2 defined by the alignment of the fine-grained muscovite. Note the occurrence of late chlorite in (d). (e–g) Highly foliated biotite alteration defining a prominent S_3 fabric. A late calcite vein cross-cuts the biotite in (g). (h, i) Coarse-grained recrystallized quartz grains. Note the presence of late calcite that is situated in the regions of recrystallized grains. (j–l) The microtextural relationship between S_2 , pre-mineralisation dolomite alteration and Zn-Pb-Ag mineralisation. Abbreviations: Ab = Albite, Bt = Biotite, Cal = Calcite, Chl = Chlorite, Crb Alt = Carbonate Alteration, Dol = Dolomite, Ms = Muscovite, Po = Pyrrhotite, Py = Pyrite, Qz = Quartz, Tmag = Titanomagnetite, PPL = Plane-Polarised Light, XPL = Cross-Polarised Light. S_x denotes the appropriate foliation.

by carbonate alteration and veining and sulphide mineralisation, constraining their paragenetic timing to post coarse-grained quartz veining and deformation (Fig. 3i–j).

4.1.6. Carbonate alteration and veining

Carbonate alteration is the most dominant stage of alteration present the drill core logged in this study (Fig. 5a–d). This stage of alteration and veining is present from 1182–1208 m in drill core NOV910301 and 460–602 m in drill core NOV900402 (Fig. 2). In order of decreasing abundance, the minerals associated with this stage of alteration include: dolomite, calcite, quartz, biotite, K-feldspar, apatite and ankerite. When less extensively developed,

this style of alteration takes the form of extensional fracture-controlled micro-veins that are locally termed “birdwing” textures (Fig. 5d). These veins are typically composed of calcite with minor quartz and range from 1–3 cm wide. With continued replacement, these veins progressively consume the siltstone and can consist of >80% of the rock as Zn-Pb-Ag mineralisation is approached, giving the rocks their ragged pseudo-breccia appearance (Fig. 5a–c). Proximal to Zn-Pb-Ag mineralisation, dolomite is the most abundant mineral (>60%) in this alteration stage, with calcite (20–30%) and quartz (10–20%) making up the remainder of the assemblage. However, calcite becomes progressively more abundant away from Zn-Pb-Ag mineralisation (Perkins, 1996). Carbonate alteration

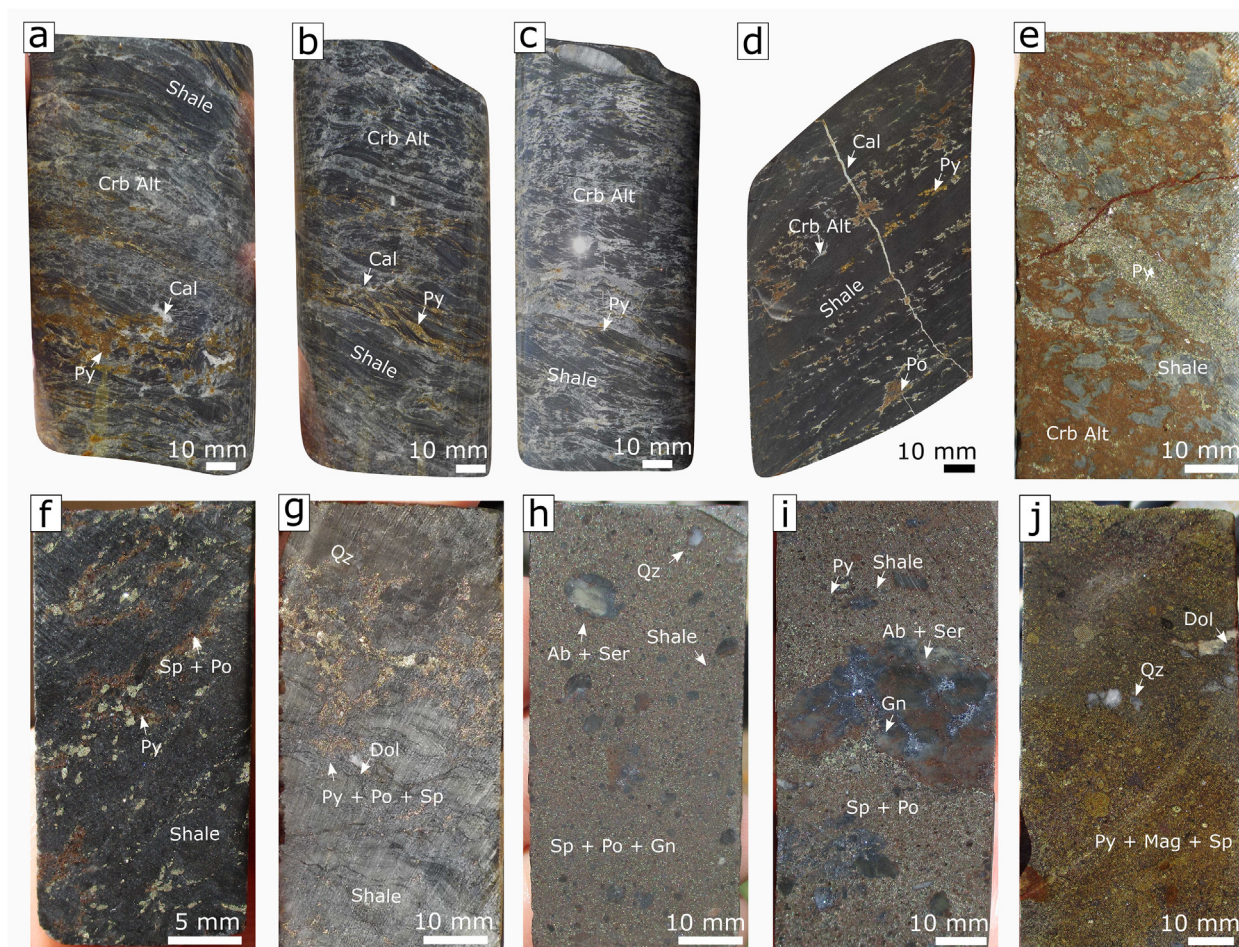


Fig. 5. Drill core samples containing pre-mineralisation alteration and the various styles of Zn-Pb-Ag mineralisation at the Mount Novit deposit. (a–c) Sections of core containing abundant pre-mineralisation carbonate alteration. Note the presence of calcite veining in (a). (d) Extensional dolomite micro-veins commonly termed “birdswings” textures. Note the replacement of the veins by pyrite and pyrrhotite. (e) Discordant and stratabound replacement of the siltstone by abundant pyrite and pyrrhotite. (f, g) Lower-grade sulphide mineralisation occurs primarily as the replacement of pre-mineralisation carbonate alteration. (h, i) High-grade Zn-Pb-Ag breccia from drill core NOV910301. Breccias are matrix-dominated with sphalerite, pyrrhotite, pyrite and galena comprising the dominant mineralogy. (j) Pyrite-magnetite-sphalerite dominated breccia from drill core NOV910301. Abbreviations: Ab = Albite, Cal = Calcite, Crb Alt = Carbonate Alteration, Dol = Dolomite, Gn = Galena, Mag = Magnetite, Py = Pyrite, Po = Pyrrhotite, Qz = Quartz, Ser = Sericite, Sp = Sphalerite.

consumes siltstone that possesses a dominant S_2 fabric (Fig. 4j–l), and the highly foliated biotite alteration, suggesting that it likely formed during late-stage D_3 deformation of the Isan Orogeny. Carbonate alteration and veining is commonly replaced by Zn-Pb-Ag mineralisation (Fig. 4j–l). Roopy chlorite seams are present in a limited number of samples where they crosscut carbonate alteration and associated veins. The chlorite veins are subsequently replaced by Zn-Pb-Ag mineralisation. Unlike coarse-grained carbonate-quartz veining, carbonate alteration and veining is fine- to medium-grained and undeformed.

4.1.7. Zn-Pb-Ag mineralisation

The northern Zn-Pb-Ag horizon occurs from 527–530 m in drill core NOV900402. In this horizon, the maximum concentration of Zn and Pb is 3.20 wt.% and 1.72 wt.% respectively. Zn-Pb-Ag mineralisation consists of coarse-grained pyrrhotite, sphalerite, pyrite, and galena (Fig. 5d–g) with rare chalcopyrite. Non-sulphide minerals in this stage include biotite and siderite (Fig. 6a–c). The sulphide minerals primary replace, and are constrained to areas of the siltstone that have undergone previous carbonate alteration and veining (Figs. 5d–g and 6a–f). In one sample, the sulphides replace a small section of the siltstone that has undergone previous albite-sericite alteration. Extending outwards from the main Zn-

Pb-Ag horizon, pyrrhotite and pyrite make up the main sulphide assemblage with minor sphalerite and rare galena. Pyrrhotite and pyrite replace carbonate alteration and veining, cross-cut biotite-rich alteration and occur in the recrystallized portions of the coarse-grained carbonate-quartz veins.

The southern Zn-Pb-Ag horizon is intersected from 1189.7–1190.9 m in drill core NOV910301. However, Zn concentrations up to 1 wt.% continue intermittently through to 1197 m. As the main Zn-Pb-Ag horizon is approached, coarse-grained pyrrhotite and pyrite are the dominant sulphides with moderate to minor concentrations of sphalerite, galena, biotite, and siderite. This assemblage primarily replaces areas of the siltstone that has undergone abundant carbonate alteration and veining (Fig. 6g–h). Zn-Pb-Ag mineralisation forms around clasts of siltstone that possess a strong S_2 foliation (Figs. 4j–l and 6g–h), showing evidence for its emplacement post- D_2 deformation. From 1189.7–1190.9 m, high grade (14.7 wt.% Zn and 9.0 wt.% Pb) Zn-Pb-Ag mineralisation occurs in a matrix-dominated polymetallic breccia (Fig. 5h–i). In this breccia, sphalerite, pyrite and pyrrhotite are the most common sulphides with less galena, arsenopyrite and rare chalcopyrite. The sulphides are often fine-grained but can range from medium- to coarse-grained. Breccia clasts range from 1 mm to 1.5 cm in diameter and are composed of fine-grained

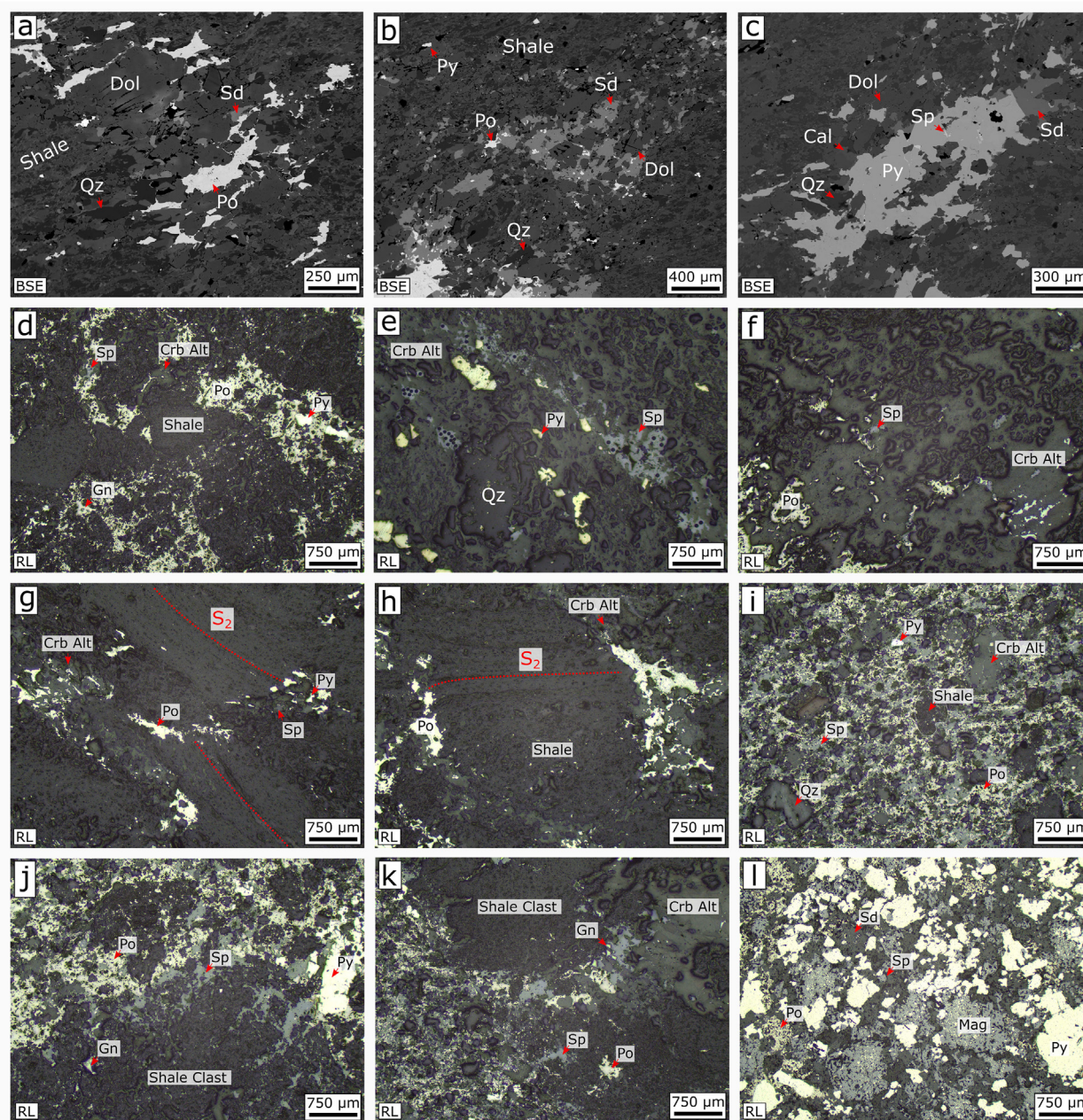


Fig. 6. Photomicrographs and SEM images showing pre-mineralisation carbonate alteration and Zn-Pb-Ag mineralisation. (a–c) Electron backscatter images showing the replacement of pre-mineralisation carbonate alteration by Zn-Pb-Ag mineralisation. (d–h) Lower-grade sulphide mineralisation occurring in the form of a pyrrhotite-pyrite-sphalerite assemblage that primarily replaces carbonate alteration. Note the relationship between S_2 , carbonate alteration and sulphide mineralisation. (i) Photomicrograph of a matrix-supported Zn-Pb-Ag breccia. Note the sharp boundaries between the clasts and the sulphide breccia. (j–k) A large siltstone clast located within the matrix-dominated Zn-Pb-Ag breccia. Note the diffuse boundary between the clast and the breccia and the presence of pre-existing carbonate alteration. (l) A photomicrograph of the magnetite-pyrite dominated breccia. Note the presence of pre-mineralisation ankerite. Abbreviations: Ank = Ankerite, Cal = Calcite, Crb Alt = Carbonate Alteration, Gn = Galena, Mag = Magnetite, Ms = Muscovite, Po = Pyrrhotite, Py = Pyrite, Qz = Quartz, Sd = Siderite, Sp = Sphalerite. S_x denotes the appropriate foliation. BSE = Electron Backscatter, RL = Reflected Light, PPL = Plane-Polarised Light.

siltstone and aggregates of calcite and quartz (Figs. 5h–i and 6i–k). Larger breccia clasts (>1 cm diameter) are made up of siltstone and have diffuse rims (Fig. 6j–k). These breccia clasts often contain discordant carbonate alteration and veining which is subsequently replaced by pyrrhotite, sphalerite and galena (Fig. 6k). Advanced albite-sericite alteration is documented in few medium-sized breccia clasts (Fig. 5h–i). Smaller breccia clasts (<0.5 cm diameter) are composed of rounded aggregates of calcite and/or quartz and possess a sharp contact with the sulphide matrix (Figs. 5h and 6i). Such breccias can be formed

from the syn-tectonic replacement of a rock or from post-depositional deformation, resulting in “durchbewegung” textures (Marshall and Gilligan, 1989). A thin section consisting of high-grade Zn-Pb-Ag mineralisation from 1190 m consists of small clasts (<0.5 cm) that possess a sharp contact with the fine-grained sulphide matrix and exhibit possible clast rotation (Fig. 6i), consistent with “durchbewegung” textures. In contrast, a sample from 1190.7 m contains some larger clasts (>1 cm) and exhibits diffuse boundaries with the surrounding sulphide matrix with no clast rotation (Fig. 6j–k), consistent with syn-tectonic

replacement. Importantly, clasts within both samples contain albite-sericite and carbonate alteration and veining (Fig. 5h–i and 6i–k), suggesting that regardless of the degree of post-emplacement deformation, Zn–Pb–Ag mineralisation must post-date these stages of alteration. At 1197 m, sulphide mineralisation is dominated by a breccia composed of fine to medium-grained pyrite and magnetite with minor pyrrhotite, sphalerite and galena (Figs. 5j and 6l). This transition is accompanied by the occurrence of siderite. In the pyrite and magnetite dominated breccia, retrograde chlorite alteration is common with chlorite laths wrapping around grains of magnetite and pyrite.

4.2. Monazite U–Pb geochronology

Both textural varieties of monazite analysed in this study were obtained from a sample collected at a depth of 468.2 m from drill core NOV900402 (Figs. 2 and 7a). Monazite U–Pb geochronology and trace element data is available in [Supplementary Data Table S1](#).

Fine-grained monazite occurs as individual grains distributed heterogeneously among the highly foliated biotite alteration (Fig. 7b–f). Individual grains are anhedral to subhedral in morphology, range from 30 to 200 μm in length and are homogeneous in Electron Backscatter Imaging (BSE) (Fig. 7c–f). Eleven analyses were excluded from the final age calculation as they possessed >10% U–Pb discordance (Fig. 8a). Successful analyses range from sub-concordant to concordant. Eighteen analyses obtained across 14 monazite grains located within the biotite-dominated mineralogy yield a mean weighted $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1527 ± 18 Ma (MSWD = 1.06) (Fig. 8a).

Monazite associated with Zn–Pb–Ag mineralisation forms within and around the cracks of recrystallized granoblastic coarse-grained quartz, occurring alongside coeval biotite, pyrite

and pyrrhotite (Fig. 7a, g–j) with minor chalcopryrite. Individual monazite grains are subhedral to anhedral and are commonly irregular in shape. Hydrothermal monazite grains are homogeneous in BSE and range from 290–800 μm in length and 110–480 μm in width (Fig. 7g–j). Three analyses were excluded from the final age calculation as they possessed >10% U–Pb discordance. Successful analyses range from sub-concordant to concordant. Forty-nine analyses obtained across 4 monazite grains associated with Zn–Pb–Ag mineralisation yield a mean weighted $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1457 ± 11 Ma (MSWD = 0.28) (Fig. 8b).

Fine-grained monazite from the biotite-rich alteration zones and coarse-grained monazite from Zn–Pb–Ag mineralisation have similar REE profiles (Fig. 8c–d). Both monazite types are characterised by a chondrite normalised LREE-enriched profile and possess only a small variation in their REE profiles (Fig. 8c–d). Despite the obvious textural differences, the two varieties of monazite also exhibit significantly different Th concentrations (Fig. 8e). Fine-grained monazite from the biotite-rich lithology possess Th concentrations ranging from 11344–22747 ppm with a median value of 16104 ppm (Fig. 8e; [Supplementary Data Table S1](#)). Monazite from Zn–Pb–Ag mineralisation possess significantly lower Th concentrations, ranging from 647–5245 ppm with a median value of 2304 ppm (Fig. 8e; [Supplementary Data Table S1](#)).

4.3. Apatite U–Pb geochronology

The apatite analysed in this study is from a sample collected at a depth of 468.2 m from drill core NOV900402 (Fig. 2). This is the same sample that contains the analysed monazite (See [Section 4.2](#)). Apatite is unzoned in BSE images and ranges from 40–100 μm in length and averages 40 μm in width (Fig. 9a–f). Apatite is located within the portion of the sample that has been significantly altered by highly foliated biotite. Apatite U–Pb geochronology and trace

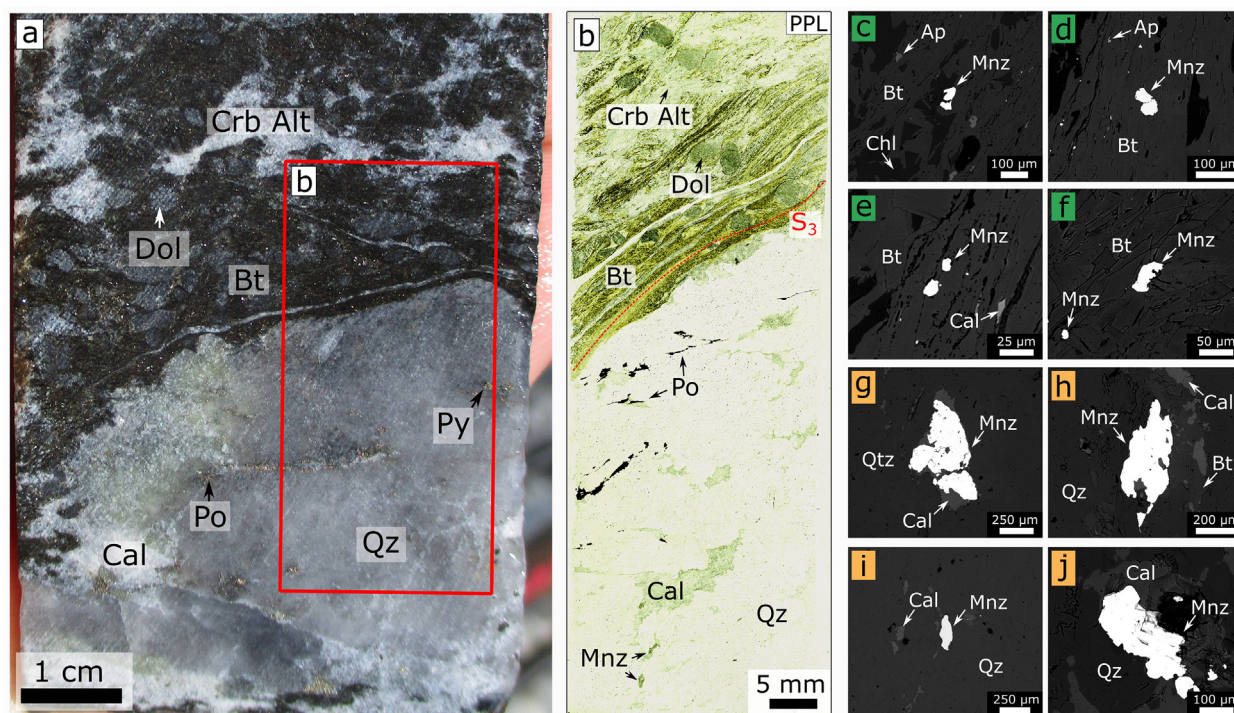


Fig. 7. Images of the monazite analyzed in this study. (a) Drill core specimen of the sample containing the two textural varieties of monazite analysed in this study. (b) Thin section image of the sample used for monazite U–Pb geochronology. Note the presence of coarse-grained monazite replacing calcite, and fine-grained monazite located within highly foliated biotite alteration. (c–f) Electron backscatter images of fine-grained monazite located in the highly foliated biotite alteration (upper portions of b). (g–j) Electron backscatter images of coarse-grained hydrothermal monazite (lower portion of b). Monazite appears to be replacing calcite and coeval with pyrrhotite and pyrite. Abbreviations: Ap = Apatite, Bt = Biotite, Cal = Calcite, Dol = Dolomite, Dol Alt = Dolomite Alteration, Mnz = Monazite, Po = Pyrrhotite, Py = Pyrite, Qz = Quartz.

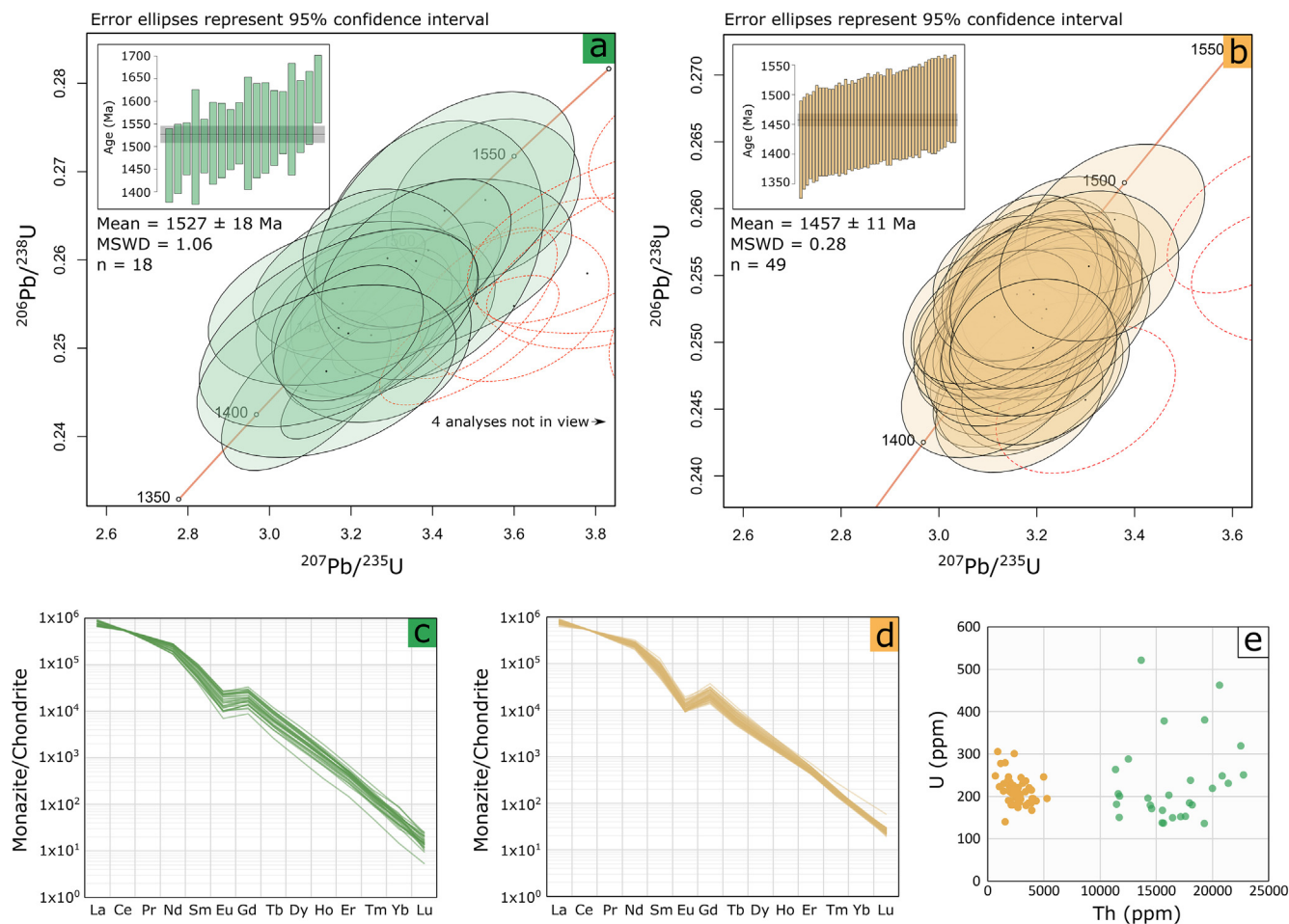


Fig. 8. Monazite U–Pb geochronology and corresponding trace element composition. Terra-Wasserburg plots and $^{207}\text{Pb}/^{206}\text{Pb}$ mean weighted ages for (a) Fine-grained monazite from the highly foliated biotite alteration and; (b) coarse-grained hydrothermal monazite coeval with Zn–Pb–Ag mineralisation. Analyses outlined in red were not included in the final age calculations. Error ellipses are shown at the 95% confidence interval. Chondrite normalized REE profile of (c) Fine-grained monazite and; (d) coarse-grained hydrothermal monazite. (e) Binary scatter plot of U vs. Th for both textural varieties of monazite analysed in this study. Hydrothermal monazite possess significant lower Th concentrations than monazite from highly foliated biotite alteration.

element data is available in [Supplementary Data Table S2](#). Fifty analyses produce a common Pb regression line with a lower intercept age of 1443 ± 29 Ma (MSWD = 1.30) and an upper intercept ($^{207}\text{Pb}/^{206}\text{Pb}$ ratio) of 0.9077 ± 0.0147 (Fig. 9g). Uranium concentrations range from 49.40–2.04 ppm with an average of 5.82 ppm. Thorium concentrations range from 334.00–0.45 ppm with an average of 3.26 ppm ([Supplementary Data Table S2](#)). Apatite shows a consistent REE pattern that is enriched in LREEs relative to MREEs and HREEs (Fig. 9h). A negative Eu anomaly is recorded in all analyses with a median Eu/Eu^* of 0.171.

4.4. Sphalerite trace elements

Sphalerite trace element data is available in [Supplementary Data Table S3](#). Sphalerite at Mount Novit contains measurable concentrations of Co, Cu, Zn, Ga, Ge, Ag, Cd, In, Sn, Sb, Tl, Pb and Bi (Fig. 10). The elements As, Se, Mo, W and Au were generally below the detection limit of the ICP-MS and are therefore not utilised in this study. The concentration of Fe in sphalerite varies from 3.20–6.70 wt.% with a median value of 4.51 wt.% (Fig. 10). Zn concentrations range from 58.54–63.35 wt.% with a median concentration of 61.91 wt.%. Cd concentrations range from 1631–2612 ppm with a median value of 2188 ppm (Fig. 10). Cu and Pb are typically present in concentrations >100 ppm and exhibit significant variations. In the analysed sphalerite, the concentration

of the elements Mn, Ag, In and Sb are between 10–100 ppm (Fig. 10). Trace elements present in concentrations <1 ppm include Co, Ga, Ge, Sn, Tl and Bi (Fig. 10). The trace element composition of sphalerite varies between the different Zn–Pb–Ag horizons. Sphalerite from drill core NOV910301 in the southern horizon contains higher concentrations of Mn, Fe, Cu, Cd, In, Co, Ga, Ge, Sn, Sb and Tl, and lower concentrations of Zn relative to sphalerite analysed from drill core NOV900402 in the northern horizon (Fig. 10).

5. Discussion

5.1. Timing of deformation, cooling and Zn–Pb mineralisation at Mount Novit

The mean weighted $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1527 ± 18 Ma obtained from monazite located within highly foliated biotite alteration (Fig. 8a) is coeval with early estimates of peak amphibolite facies metamorphism and D_2 deformation in the Mount Novit area at 1530 ± 7 Ma (Connors and Page, 1995). This estimate was produced from U–Pb zircon geochronology performed on a pre to syn- D_2 pegmatite that contained two age populations at 1532 ± 7 Ma and 1565 ± 5 Ma respectively (U–Pb zircon; Connors and Page, 1995). The older age population was interpreted to represent the age of inherited zircons while the younger age population was interpreted to represent the age of the pegmatite emplacement

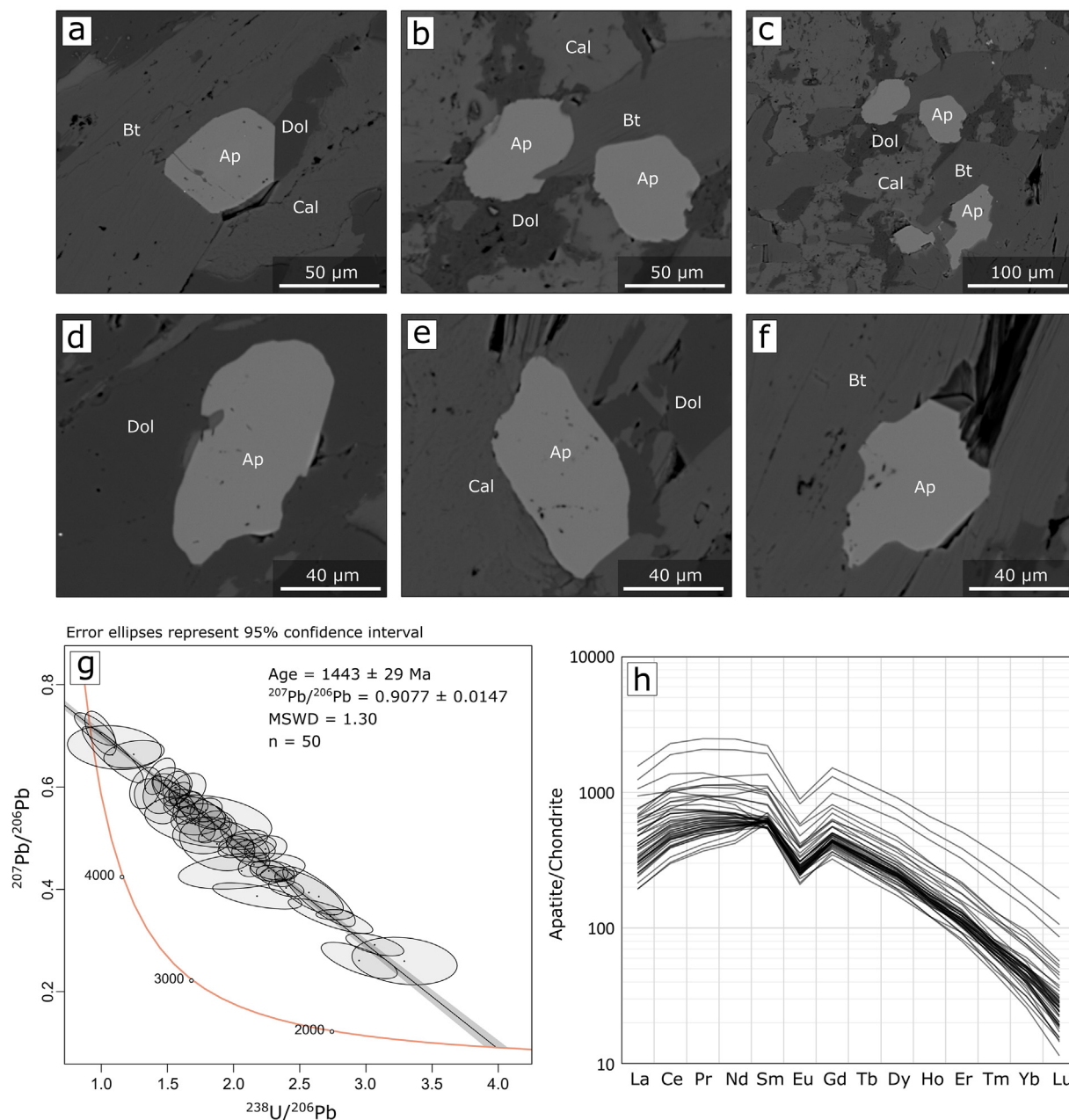


Fig. 9. Apatite textures and corresponding U–Pb geochronology and trace elements. (a–f) Electron backscatter images of the apatite analysed in this study. (g) A Terra–Wasserburg plot showing the results from apatite U–Pb geochronology. Error ellipses are shown at the 95% confidence interval. (h) A chondrite normalized REE profile of the apatite analysed in this study. Abbreviations: Dol = Dolomite, Cal = Calcite, Bt = Biotite, Ap = Apatite.

(Connors and Page, 1995). However, this age is inconsistent with the currently accepted age of peak metamorphic conditions in the Western Fold Belt of the Mount Isa Inlier at ca. 1570 Ma (U–Pb monazite; Hand and Rubatto, 2002). We re-interpret the data of Connors and Page (1995) to suggest that the older age peak of 1565 ± 5 Ma likely represents the age of the syn- D_2 pegmatite intrusion with the younger 1532 ± 7 Ma age reflecting subsequent zircon growth during retrograde metamorphic conditions associated with D_3 deformation of the Isan Orogeny. Biotite from Fe-rich alteration zones contemporaneous with the formation of the silica-dolomite alteration at Mount Isa yields $^{40}\text{Ar}/^{39}\text{Ar}$ ages from 1523–1516 Ma (Perkins et al., 1999). Biotite from the Andersons Lode U deposit, located approximately 15 km east of Mount Isa produces a similar $^{40}\text{Ar}/^{39}\text{Ar}$ age of ca. 1534 Ma (Perkins et al.,

1999). These ages were interpreted to represent the transition from peak regional metamorphism to later metamorphic-hydrothermal activity (Perkins et al., 1999). More recently, a $^{40}\text{Ar}/^{39}\text{Ar}$ hornblende age of 1523 ± 7 Ma has been produced from a foliated granite and a $^{40}\text{Ar}/^{39}\text{Ar}$ amphibole age of 1505 ± 6 Ma has been produced from an amphibolite in the Sybella region (Li et al., 2020). These ages correspond with the cooling temperatures of 532 ± 54 °C and 507 ± 53 °C respectively (Li et al., 2020). Consistent with previous studies, the U–Pb monazite age of 1527 ± 18 Ma is interpreted to represent the formation of the highly foliated biotite-rich alteration assemblage and coeval retrograde metamorphism during D_3 deformation of the Isan Orogeny.

Apatite U–Pb geochronology is commonly used as a mid-range geothermometer with a closure temperature ranging between

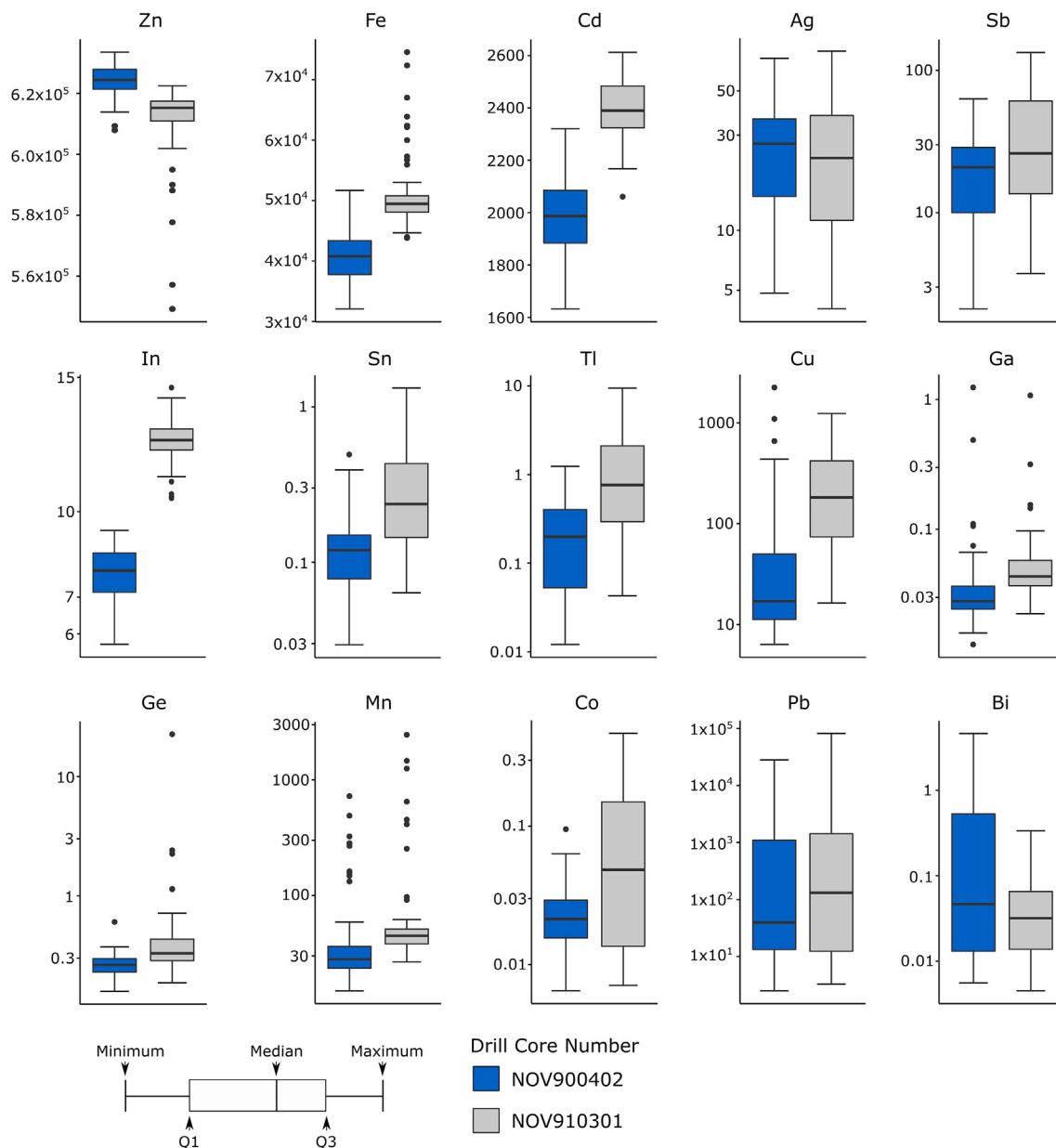


Fig. 10. Box and whiskers plots displaying the trace element composition of sphalerite from the Mount Novit Zn-Pb-Ag deposit. All results are presented in parts per million (ppm).

375–550 °C (Cochrane et al., 2014; Chew and Spikings, 2015). The low U–Pb closure temperature of apatite allows cooling and exhumation ages to be accurately determined (e.g. Hall et al., 2018; Jepson et al., 2021). The closure temperature of apatite is dependent upon grain size and the rate of cooling (Cherniak et al., 1991; Cochrane et al., 2014). Coarse-grained apatite crystals (10–20 mm) generally possess a high closure temperature of ~620 °C (Krogstad and Walker, 1994) while finer-grained apatite (10–50 μm) generally possess a lower closure temperature of ~375–550 °C (Cherniak et al., 1991; Chamberlain and Bowring, 2001; Cochrane et al., 2014; Chew and Spikings, 2015). As the analysed apatite from Mount Novit is relatively fine-grained (averaging 40 μm in width; Fig. 9a–f), the lower intercept age of 1443 ± 29 Ma is interpreted to reflect cooling of the apatite below the lower closure temperature estimates of ~375 °C (Cherniak et al., 1991; Chamberlain and Bowring, 2001). The U–Pb apatite age presented in this study broadly correlates with the $^{40}\text{Ar}/^{39}\text{Ar}$ age of

1463 ± 3 Ma produced from metamorphic hornblende at Mica Creek located approximately 5 km north of the Mount Novit deposit (Perkins et al., 1999). The ca. 1460 Ma age was interpreted to represent cooling (<500 °C) of the Mica Creek area as a result of major movement along the Mount Isa Fault (Perkins et al., 1999). This movement resulted in the amphibolite-facies rocks on the western side of the Mount Isa Fault being thrust up against the lower greenschist facies rocks on the eastern side of the Mount Isa Fault (Perkins et al., 1999). The U–Pb apatite age of 1443 ± 29 Ma also correlates with the $^{40}\text{Ar}/^{39}\text{Ar}$ hornblende age of 1439 ± 5 Ma from the Eastern Creek Volcanics north of the Mount Novit area (Spikings et al., 2002) and a biotite $^{40}\text{Ar}/^{39}\text{Ar}$ age of 1444 ± 9 Ma obtained in the northern region of the Sybella Granite (Spikings et al., 2002). Similar to Perkins et al. (1999), these ages were interpreted to represent an increased rate of exhumation of between 4.5–10 km of crust from 1440–1370 Ma (Spikings et al., 2002).

Petrological evidence provided in this study indicates that Zn-Pb-Ag mineralisation at Mount Novit overprints the dominant S_2 foliation found in the metasedimentary host rocks (Fig. 6a–c, i–j). Further evidence presented in this study shows the monazite dated at 1457 ± 11 Ma replaces calcite and grew synchronously with pyrrhotite (Fig. 7b, g–j) and chalcopryrite associated with Zn-Pb-Ag mineralisation. Based on the textural constraints presented in this study, this age is interpreted to represent the best estimate regarding the timing of Zn-Pb-Ag mineralisation in the Mount Novit area. Due to the low availability of datable minerals at the nearby George Fisher, Hilton and Mount Isa Zn-Pb-Ag deposits, no absolute geochronology has been performed on these deposits. Pb-model ages have been constructed for the region and generally favour a Zn-Pb-Ag mineralisation age of ca. 1655 Ma (Carr et al., 2001, 2004). This is in contrast to the structural and textural studies on the individual Zn-Pb-Ag deposits which favour their formation during late- D_3 deformation of the Isan Orogeny (Perkins, 1997; Perkins and Bell, 1998; Davis, 2004; Murphy, 2004). There has been increasing debate regarding the co-genesis of Cu and Pb-Zn-Ag mineralisation across the Mount Isa region (Perkins, 1997; Perkins and Bell, 1998; Davis, 2004; Cave et al., 2020a). While there is a lack of absolute geochronological constraints, it is generally accepted that Cu mineralisation at the Mount Isa, Hilton, George Fisher and Mammoth deposits was emplaced syn- to late- D_3 deformation of the Isan Orogeny (Perkins, 1984; Swager, 1985; Bell et al., 1988; Valenta, 1994b; Clark, 2003; Chapman, 2004). Biotite from a pyrrhotite vein associated with Cu mineralisation produced an $^{40}\text{Ar}/^{39}\text{Ar}$ age of ca. 1505 Ma (Perkins et al., 1999). Pooled whole-rock Rb–Sr data from multiple tuff marker beds at Mount Isa produced an imprecise age of 1482 ± 52 Ma, which was interpreted to represent the closure of the Rb–Sr system and the cessation of metamorphism (Page, 1981). Perkins (1984) re-examined the Rb–Sr age and suggesting that it represents the development of a D_3 cleavage and therefore the maximum age of Cu mineralisation. In the Mount Novit area, a post- D_3 pegmatite has also been dated at 1480 ± 14 Ma (Connors and Page, 1995), providing evidence to suggest the presence of retrograde metamorphic-hydrothermal fluids. Although the absolute significance of the 1457 ± 11 Ma mineralisation age is yet to be understood across the Mount Isa Inlier, the current work produces the first age of Zn-Pb-Ag mineralisation in the Mount Isa region using conventional geochronological methods. The resulting age provides strong evidence to support the presence of post to late- D_3 Zn-Pb-Ag mineralisation in the Mount Isa region.

5.2. Nature of the Zn-Pb-Ag-bearing fluids

The trace element content of sphalerite reflects the trace element content of the fluid source and its co-crystallizing mineral assemblage (Cook et al., 2009; Belissont et al., 2014; Frenzel et al., 2016; Cave et al., 2020b). The elements Co, Ag, Sb, Tl, Pb and Bi preferentially substitute into co-crystallizing pyrite, pyrrhotite and galena, which can greatly affect the content of these elements in co-genetic sphalerite (George et al., 2016; Cave et al., 2020b). As the pyrrhotite and pyrite are the most dominant sulphides present at Mount Novit, the low concentration of Co and Tl likely reflects the abundant co-crystallization of these minerals.

Previous studies indicate that the concentration of the trace elements Ga, Ge and In within sphalerite systematically change with precipitation temperature and can broadly reflect the genetic characteristics of the fluid from which they precipitated (Belissont et al., 2014; Frenzel et al., 2016; Bauer et al., 2019a). Sphalerite with high concentrations of In is typically associated with high-temperature deposits with a magmatic-hydrothermal affiliation (Frenzel et al., 2016; Bauer et al., 2019a) or a high abundance of Cu within the precipitating fluid, promoting the coupled substitu-

tion $2\text{Zn}^{2+} \rightarrow \text{Cu}^+ + \text{In}^{3+}$ (Bauer et al., 2019b). Sphalerite with higher Ga and Ge concentrations is typical of low-temperature deposits with no obvious magmatic-hydrothermal affiliation (Frenzel et al., 2016; Bauer et al., 2019a). Based on the studies cited above, the low concentrations of Ga, Ge, and the relative enrichment of In within sphalerite from Mount Novit favours a relatively hot fluid source with a magmatic-hydrothermal affinity. The relatively low concentration of Ge and Ga relative to In is consistent with the nearby Mount Isa and Hilton Zn-Pb-Ag deposits (Cave et al., 2020a,b). While a magmatic fluid source is plausible, there is no magmatism in the Mount Isa region currently recognised at ~ 1460 Ma, making a hydrothermal-magmatic origin of the Zn-Pb-Ag mineralising fluid improbable. Alternatively, a high concentration of In relative to Ga and Ge could be achieved if the metals were leached from an underlying basement unit that possesses a similar abundance of these elements. It is suggested that further research is required to understand the source of the Zn-Pb-Ag mineralising fluid at the Mount Novit deposit.

The concentration of the elements Ga, Ge, In, Mn and Fe in sphalerite adequately correlate with the precipitation temperature of sphalerite, allowing their concentrations to be used as a geothermometer (Frenzel et al., 2016). Combined, sphalerite from NOV900402 and NOV10301 produced a range of temperatures from $282\text{--}389^\circ\text{C}$ with a median value of $345 \pm 52^\circ\text{C}$ (Fig. 11). This is directly comparable to the estimated sphalerite formation temperature of $343 \pm 44^\circ\text{C}$ at the Mount Isa Zn-Pb-Ag deposit (Fig. 10; Cave et al., 2020a). This estimate is slightly hotter than the median sphalerite formation temperature of $311 \pm 44^\circ\text{C}$ at the George Fisher Zn-Pb-Ag deposit (GGIMFis geothermometer, unpublished data) and the median sphalerite formation temperature of $262 \pm 24^\circ\text{C}$ at the Hilton Zn-Pb-Ag deposit (GGIMFis geothermometer, Cave et al., 2020b). These temperature estimates are also consistent with the estimated formation temperature of silica-dolomite alteration ($270\text{--}350^\circ\text{C}$; Heinrich et al., 1989) and Cu mineralisation ($272\text{--}323^\circ\text{C}$; Heinrich et al., 1989) at the Mount Isa deposit (Fig. 10). In the Mount Novit area, the estimated temperature of sphalerite precipitation is consistent with the estimated temperature of late- to post- D_3 deformation from $450\text{--}300^\circ\text{C}$ based on mineral stability reactions (Connors and Lister, 1995). This estimate also overlaps with the interpreted closure

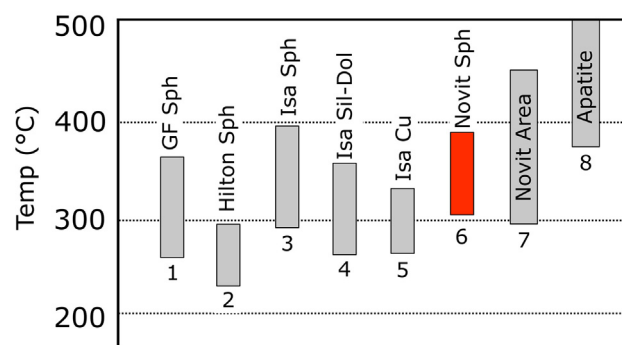
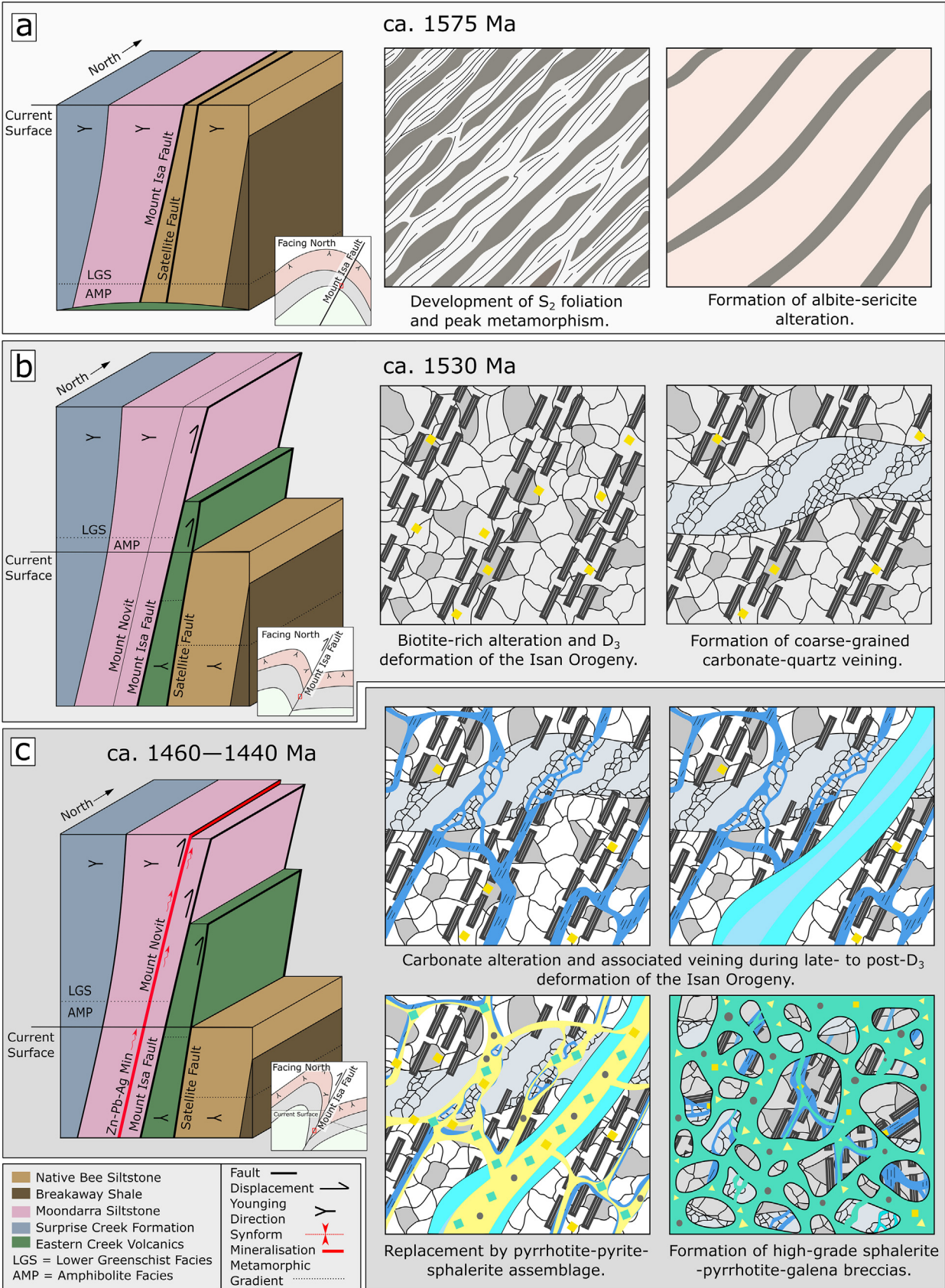


Fig. 11. Temperature constraints for nearby Cu and Zn-Pb-Ag deposits and regional metamorphism. Estimated precipitation temperature of sphalerite from (1) George Fisher (GF; GGIMFis Geothermometer, unpublished data) and (2) Hilton (data sourced from Cave et al. (2020b)). (3) Temperature of sphalerite precipitation from Mount Isa (Cave et al., 2020a). (4) Estimated temperature of silica-dolomite alteration at Mount Isa (Heinrich et al., 1989). (5) Estimated temperature of Cu mineralisation at Mount Isa (Heinrich et al., 1989). (6) Temperature of sphalerite precipitation from Mount Novit using GGIMF is geothermometer. (7) Estimated temperature of the Mount Novit region from late- to post- D_3 deformation (Connors and Page, 1995). (8) Estimated temperature of Pb diffusion in apatite (Cochrane et al., 2014).



temperature of apatite ($\sim 375^\circ\text{C}$ at 1443 ± 29 Ma) presented in this study.

5.3. A metallogenic model for the Mount Novit region

The Mount Novit area experienced upper amphibolite facies metamorphism during D_2 of the Isan Orogeny, resulting in the dominant S_2 fabric present in the samples from this study (Fig. 12a). Although the age of this event is poorly defined, the most recent estimates place this event at ca. 1575 Ma (Hand and Rubatto, 2002; Duncan et al., 2006). Albite-sericite alteration at Mount Novit is constrained to either syn- or pre- D_2 of the Isan Orogeny (Fig. 12a). Comparably, albite-bearing “buff” alteration at Mount Isa is also interpreted to have formed during D_2 deformation of the Isan Orogeny (Perkins, 1997).

Monazite from the highly foliated biotite-rich alteration produces a $^{207}\text{Pb}/^{206}\text{Pb}$ mean weighted age of 1527 ± 18 Ma (Fig. 8a). This is consistent with this stage of alteration forming synchronously with early D_3 deformation of the Isan Orogeny (Fig. 12b) and coeval movement along the Mount Isa Fault. This age also correlates with the $^{40}\text{Ar}/^{39}\text{Ar}$ age of 1523–1516 Ma regarding phyllosilicate development at the Mount Isa mine, the $^{40}\text{Ar}/^{39}\text{Ar}$ biotite age of ca. 1534 Ma acquired from the Andersons Lode U deposit and the $^{40}\text{Ar}/^{39}\text{Ar}$ ages 1523 ± 7 Ma and 1505 ± 6 Ma obtained from micas in schistose units in the Sybella region (Li et al., 2020). It is interpreted that coarse-grained quartz-carbonate veins formed during this deformation event, where they experienced ongoing deformation after their emplacement, resulting in the formation of boudins and recrystallized granoblastic quartz (Fig. 12b).

Significant movement along the Mount Isa and Satellite Faults occurred during late- to post- D_3 deformation of the Isan Orogeny (Connors et al., 1992). Although there is a lack of younging indicators, continuation with the Crystallina Block indicates the section of the Eastern Creek Volcanics between the Mount Isa and Satellite Fault is west younging (Figs. 2 and 11b). As the Mount Novit Zn-Pb-Ag horizon is parallel to the adjacent Mount Isa and Satellite Fault Zones (Fig. 2), a similar history of movement and dilation is inferred. Movement along the Mount Isa and Satellite Fault Zones during D_3 deformation thrust the eastward younging (amphibolite facies) Moondarra Siltstone up against the westward younging (lower greenschist facies) Eastern Creek Volcanics and Native Bee Siltstone (Fig. 12c). Movement along these faults facilitated the transport of fluids into the Mount Novit Zn-Pb-Ag horizon, resulting in the locally abundant carbonate alteration and veining (Figs. 5a–c and 12c). Displacement along the Mount Isa Fault continued during late- to post- D_3 , with F_4 and S_4 folds and fabrics recognised throughout the Mount Novit area (Connors and Lister, 1995). Apatite U–Pb geochronology presented in this study is interpreted to record this (late- D_3 to D_4) cooling event ($\sim 375^\circ\text{C}$) at 1443 ± 29 Ma. This age is consistent with previous $^{40}\text{Ar}/^{39}\text{Ar}$ geochronology, which has previously dated an uplift event at

1463 ± 3 Ma (Perkins et al., 1999). Movement along the Mount Isa and Satellite Fault Zones is interpreted to have driven fluid flow and facilitated the influx of Zn-Pb-Ag mineralisation, which is dated by monazite U–Pb geochronology at 1457 ± 11 Ma (Fig. 8b). The sphalerite GGIMFis geothermometer (Frenzel et al., 2016) suggests that the Zn-Pb-Ag mineralisation was relatively hot ($\sim 350^\circ\text{C}$) upon placement, consistent with the cooling age and temperature derived from apatite U–Pb geochronology.

Zn-Pb-Ag mineralisation in the northern horizon, and lower grade (<1 wt.% Zn + Pb) mineralisation in the southern horizon is primarily constrained to areas of the siltstone that has been affected by carbonate alteration and veining. This suggests that this alteration stage provided a favourable pathway that was exploited by the subsequent Zn-Pb-Ag-bearing fluid. In this case, the dissolution and replacement of the pre-existing dolomitic alteration would have resulted in a localised increase of pH and assisted in the precipitation of sulphide mineralisation (Reed and Palandri, 2006; Liu et al., 2021). In high-grade Zn-Pb-Ag breccias, such as 1189.7–1190.9 m in drill core NOV910301, mild fracturing and brecciation associated with fault movement is interpreted to have provided a suitable pathway for the subsequent mineralising fluid. Aided by the dissolution of carbonate, the precipitation of Zn-Pb-Ag mineralisation is interpreted to have been primarily driven by cooling ($\sim 375^\circ\text{C}$) in the Mount Novit area as a consequence of uplift along the adjacent Mount Isa Fault (Fig. 12c).

Although the geochronology presented in this study is consistent with previous geochronological results and interpretations, it is limited by the use of a single sample for monazite and apatite U–Pb geochronology. We suggest that additional work is required to determine the full significance of the 1443 ± 29 Ma apatite and 1457 ± 11 Ma monazite ages presented in this study.

6. Conclusions

- Monazite from highly foliated biotite-alteration in the Mount Novit area of the Mount Isa Inlier, Australia, yields a $^{207}\text{Pb}/^{206}\text{Pb}$ mean weighted age of 1527 ± 18 Ma (MSWD = 1.06). This age is consistent with its formation during D_3 deformation of the Isan Orogeny.
- Apatite from highly foliated biotite-alteration produces a lower intercept age of 1443 ± 29 Ma (MSWD = 1.30). This is interpreted to represent the age of uplift along the Mount Isa Fault and consequent cooling below $\sim 375^\circ\text{C}$ in the Mount Novit area.
- Monazite coeval with Zn-Pb-Ag mineralisation produces a $^{207}\text{Pb}/^{206}\text{Pb}$ age of 1457 ± 11 Ma (MSWD = 0.28).
- Sphalerite from Mount Novit contains low concentrations of Ge and Ga (<1 ppm) and relatively high concentrations of In (5 to >10 ppm). This is interpreted to reflect the leaching of these metals from a basement unit with similar elemental quantities. The GGIMFis geothermometer produces an estimated sphalerite formation temperature of $\sim 350^\circ\text{C}$.

Fig. 12. Conceptual model of the formation of the Mount Novit Zn-Pb-Ag deposit following the structural evolution outlined by Connors (1992) and Gordon (2004). (a) Early metamorphism was coeval with the development of regional-scale D_2 deformation that produced a large-scale F_2 anticline in the Mount Novit region (a). This event is coeval with the formation of albite-sericite alteration and the S_2 metamorphic fabric. (b) Early D_3 deformation consisted of west over east movement and resulted in the reverse reactivation of the Mount Isa and Satellite Faults. Reverse reactivation of the Mount Isa Fault overturned the eastern limb of the large-scale F_2 anticline relative to the units on the eastern side of the Mount Isa Fault. This event displaces the eastward younging (amphibolite facies) Moondarra Siltstone up against the westward younging (lower greenschist facies) Eastern Creek Volcanics and Native Bee Siltstone. Early D_3 deformation is interpreted to be coeval with significant biotite alteration and coarse-grained calcite-quartz veining. (c) Continued thrusting along the Mount Isa Fault from late D_3 facilitated pre-mineralisation carbonate alteration and veining in the Mount Novit horizon. This alteration stage is subsequently replaced by Zn-Pb-Ag mineralisation. Uplift associated with the continued thrusting along the Mount Isa Fault, during late D_3 or possibly during D_4 as described by Connors and Lister (1995) resulted in the cooling of the Mount Novit block below $\sim 375^\circ\text{C}$ at ca. 1440 Ma. Note the inserts depicting the larger scale structure of the Mount Novit region with the approximate location of the block model shown in Fig. 12a–c outlined in red.

- Based on the evidence from this study it is suggested that uplift along the Mount Isa Fault at ca. 1460–1440 Ma facilitated the transport of Zn-Pb-Ag mineralising fluids along the Mount Novit horizon, with precipitation primarily resulting from the cooling of the Mount Novit area below ~375 °C.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to thank Peter Rea and the rest of the Mount Isa Mines Exploration Team for their assistance and hospitality during fieldwork and for organising the viewing and sampling of the Mount Novit core. We would like to thank Roger Taylor and Tim Bell for their assistance and feedback with the petrology completed in this study. We would also like to thank Karin Barovich for her assistance with drafting this manuscript. Sarah Gilbert and Aoife McFadden from Adelaide Microscopy are thanked for their assistance with analytical procedures. B.C. is supported by the Australian Government Research Training Program. This study was funded by Mount Isa Mines—a Glencore Company and forms part of the Mount Isa Research for Geology and Exploration (MIRGE) project.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gsf.2021.101347>.

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